COHOMOLOGICAL DIMENSION OF A ONE-DIMENSIONAL AFFINE SCHEME OVER A FIELD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; the final numerical inference from the vanishing region of the spectral sequence to the bound cd(X) at most 1 + cd(K) is left implicit by the author and is spelled out in this card's core_obligations without altering the counted text.

Editorial note. The label lemma-cd-curve-over-field occurs in no other staged Stacks chapter, so no cross-chapter restatement had to be weighed for this card.

Editorial note. The single labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer to the same chapter.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Cohomological dimension cd(X) of a scheme. Same chapter, the definition of the cohomological dimension as the supremum of the degrees in which torsion sheaves can have nonzero cohomology. Supplies the quantity bounded in the statement.
- Leray spectral sequence of a morphism of schemes. Same chapter, the Leray spectral sequence E_2^{p,q} = H^p(S, R^q f_*F) converging to H^{p+q}(X, F). Supplies the spectral sequence with which the proof reduces the bound to the vanishing of the higher direct images.
- Vanishing of cohomology of torsion sheaves on an affine curve in degrees at least two. Same chapter, the theorem that H^i of a torsion sheaf on a one-dimensional affine scheme over a field vanishes for i at least 2. Supplies the vanishing of the stalks of R^q f_*F in degrees at least 2, which is the input to the spectral sequence argument.

Source reference. Original theorem 03SC: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 15656–15656. Same-chapter navigation pointer invoked at proof line 19668-19669, where the author uses that the cohomology of an affine curve vanishes in degrees at least two. The pointer only resolves that reference for the reader; the cited theorem is not reproduced inside the counted span, and no bibliography citation is involved.

Lemma 0.1. *Let K be a field. Let X be a 1-dimensional affine scheme of finite type over K. Then $cd(X) \leq 1 + cd(K)$.*

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Proof. Let $\mathcal{F}$ be an abelian torsion sheaf on $X_{étale}$. Consider the Leray spectral sequence for the morphism $f : X \to \mathrm{Spec}(K)$. We obtain

$$E_2^{p,q} = H^p(\mathrm{Spec}(K), R^q f_* \mathcal{F})$$

converging to $H^{p+q}_{étale}(X, \mathcal{F})$. The stalk of $R^q f_* \mathcal{F}$ at a geometric point $\mathrm{Spec}(\overline{K}) \to \mathrm{Spec}(K)$ is the cohomology of the pullback of $\mathcal{F}$ to $X_{\overline{K}}$. Hence it vanishes in degrees ≥ 2 by Theorem 03SC. $\square$